Mini Project Report

on

Automated Diagram Digitization

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In partial fulfilment of the requirements for the award of the degree of Bachelor of Technology in Computer Science and Engineering.

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CERTIFICATE

Certified that this is a Mini Project Report titled

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DECLARATION

We, Nema Fathima, Nidha Rahma, Rithuparna J B, Rose Maria and Sana Noushad, hereby declare that the project titled **"Automated Diagram Digitization"** is a bona fide work carried out by us during the year 2026, under the guidance of **Ms. Swaiba Nasmi**, Project Guide, Division of Computer Science and Engineering, School of Engineering, CUSAT. We further declare that this project has not been submitted, either in part or full, for the award of any degree, diploma, associateship, fellowship, or any other academic recognition in any university or institution.

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ABSTRACT

Automated Diagram Digitization is a modern web-based application designed to convert hand-drawn or printed flowchart images into interactive, editable digital diagrams. This project addresses the critical challenge of manually recreating diagrams by leveraging artificial intelligence to automatically recognize shapes, extract text, and detect connections between nodes. The system utilizes Google's Gemini AI Vision model to analyze uploaded flowchart images and identify four basic shape types: process (rectangle), decision (diamond), start/end (oval), and input/output (parallelogram). The extracted information is structured into a JSON format compatible with React Flow, enabling users to interact with and modify their diagrams through an intuitive drag-and-drop interface. The system features three input methods: image upload, text-to-flowchart generation from natural language descriptions, and JSON import for existing flow data. Key functionalities include an interactive editor with undo/redo, auto-layout for clean alignment, edge labeling via double-click, text styling (bold, italic, font size), node/edge color customization, and multiple export options (PNG, SVG, JSON). Built with FastAPI on the backend, PostgreSQL for persistent storage, and React with Vite on the frontend, Automated Diagram Digitization demonstrates a production-ready solution for flowchart automation. The platform implements robust error handling for AI responses, custom prompt engineering for accurate shape recognition, and comprehensive API endpoints for all operations. With features like theme switching (dark/light mode), node resizing with text adaptation, and a drag-drop shape palette, the system provides a seamless experience for students, educators, software developers, and business professionals who need to digitize and edit flowcharts efficiently. **Keywords:** Diagram Digitization, Optical Graph Recognition, Gemini AI, React Flow, FastAPI, Flowchart Automation, Computer Vision, Natural Language Processing

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Chapter 1

Introduction

In today's digital era, the need to convert hand-drawn or printed diagrams into editable digital formats has become increasingly important across various domains including education, software development, and business process modeling. Flowcharts, being fundamental tools for representing algorithms, processes, and workflows, are frequently created on paper or whiteboards during brainstorming sessions, classroom teachings, and initial design phases. However, these hand-drawn diagrams often remain in physical form, limiting their usability, shareability, and editability.

Traditional methods of digitizing flowcharts involve manual redrawing using diagramming tools like Microsoft Visio, Lucidchart, or draw.io—a time-consuming and error-prone process. This creates a significant gap between the creative, free-form sketching process and the structured, digital representation required for documentation and further processing. Recent advances in artificial intelligence, particularly in computer vision and large language models, have opened new possibilities for automated diagram recognition, enabling the development of intelligent systems that can understand and reconstruct diagrammatic information automatically.

1.1 Problem Statement

The existing diagram digitization ecosystem suffers from the following key problems:

- **Manual Redrawing Required**: Users must manually recreate diagrams using tools like Visio or draw.io, a process that can take 15-30 minutes for moderately complex flowcharts, wasting time and introducing errors.
- **Lack of Automated Recognition**: Current systems cannot automatically identify shapes, text, and connections from images of hand-drawn or printed flowcharts, requiring complete manual intervention.
- **Text Extraction Challenges**: OCR tools struggle with handwritten text and cannot preserve line breaks or formatting within shapes, resulting in loss of readability and meaning.
- **Connection Detection Gap**: Existing systems fail to recognize the directional relationships between shapes, requiring manual connection drawing, which is particularly problematic for complex flowcharts with multiple decision paths.
- **Limited Input Methods**: Most tools only support manual creation, lacking support for image upload, text-to-flowchart generation, or JSON import for existing flow data.
- **No Intelligent Editing Features**: Absence of auto-layout, undo/redo, text styling, and edge labeling capabilities in automated tools, limiting post-recognition editing flexibility.

- **Export Limitations**: Lack of multiple export formats (PNG, SVG, JSON) for different use cases such as documentation, web embedding, or data portability.

1.2 Objectives

The primary objectives of the Automated Diagram Digitization project are:

1. **AI-Powered Recognition**: Develop a system capable of recognizing flowchart shapes, text, and connections from images using Google's Gemini AI with custom prompt engineering for accurate shape identification.
2. **Multiple Input Methods**: Provide three input modes—image upload, text description for algorithm-to-flowchart conversion, and JSON import—for maximum user flexibility and workflow integration.
3. **Interactive Diagram Editor**: Create an intuitive drag-drop editor with undo/redo functionality, auto-layout for clean alignment, and comprehensive styling options including node/edge color customization.
4. **Text Customization**: Enable text formatting including bold, italic, and font size adjustments for node content to enhance readability and presentation quality.
5. **Edge Labeling**: Implement double-click edge editing for adding and modifying connection labels, supporting decision branches with "Yes/No" labels and proper styling.
6. **Export Functionality**: Support multiple export formats including PNG for sharing, SVG for vector graphics, and JSON for data portability and integration with other tools.
7. **Theme Support**: Implement dark/light mode toggle with persistent user preference stored in localStorage to reduce eye strain and accommodate user preferences.
8. **Persistent Storage**: Enable saving and retrieving flowcharts using PostgreSQL database with UUID-based identification and timestamp tracking for audit purposes.

1.3 Scope of the Project

The scope of Automated Diagram Digitization encompasses the following aspects:

1.3.1 Functional Scope

- **Shape Recognition**: Identification of four basic flowchart shapes—process (rectangle), decision (diamond), start/end (oval), and input/output (parallelogram)—from uploaded images.
- **Text Extraction**: Accurate extraction of text from shapes with proper line break preservation using \n characters for multi-line labels.

- **Connection Detection**: Automatic detection of edges between nodes with directional arrows and appropriate handle assignments (top, bottom, left, right).
- **Image Input**: Support for PNG, JPG, and WebP image formats with drag-drop interface and file size validation.
- **Text-to-Flowchart**: Natural language processing for converting algorithm descriptions into structured flowcharts with proper node types and connections.
- **JSON Import/Export**: Ability to import existing flowchart JSON data for editing and export current flow state for data portability.
- **Interactive Editor**: Drag-drop from shape palette, node resizing with text adaptation, undo/redo with keyboard shortcuts, and auto-layout for alignment.
- **Styling Features**: Node/edge color customization, text styling (bold, italic, font size), and edge labeling with background styling for readability.

1.3.2 Technical Scope

- **Backend Development**: Implementation using FastAPI (Python) with asynchronous endpoints for image analysis, text-to-flowchart generation, and database operations.
- **Frontend Development**: Development using React 18 with Vite for optimal performance, React Flow for diagram rendering, and custom node components for each shape type.
- **AI Integration**: Integration with Google Gemini AI API using custom system prompts for accurate shape recognition and text-to-flowchart conversion.
- **Database Design**: PostgreSQL database with JSONB support for storing complete React Flow state, UUID primary keys, and timestamp tracking.
- **Export Implementation**: PNG export using html-to-image library, SVG export using React Flow's SVG renderer, and JSON export with proper serialization.
- **API Design**: RESTful API endpoints for all operations including image analysis, text generation, and flowchart CRUD operations.

1.3.3 Limitations and Exclusions

The project does not cover recognition of complex BPMN symbols, swimlanes, UML diagrams, or real-time video processing. It focuses specifically on basic flowchart shapes commonly used in algorithm representation and process documentation.

Chapter 2

Literature Review

The integration of artificial intelligence into document processing and diagram recognition has seen rapid development in recent years, particularly in enhancing automated understanding of visual information. Researchers and industry practitioners have explored various approaches to digitizing hand-drawn diagrams, recognizing shapes, extracting text, and understanding structural relationships. This chapter reviews recent literature related to computer vision for diagram recognition, optical character recognition for handwritten text, AI-powered code and flowchart generation, and interactive diagramming tools.

The technologies used in diagram digitization include convolutional neural networks (CNNs) for shape detection, optical character recognition (OCR) engines for text extraction, large language models (LLMs) for understanding diagram context, and interactive web frameworks for diagram editing. However, the literature also identifies limitations such as accuracy challenges with hand-drawn diagrams, difficulty in detecting connection paths, poor handling of overlapping elements, and the lack of integrated solutions combining recognition with editing capabilities.

2.1 Computer Vision for Diagram Recognition

This study presents a unified framework that merges text recognition with generative intelligence for automated document processing, focusing on accurate extraction, summarization, and contextual interpretation of visual data. The research incorporates preprocessing techniques like noise reduction and layout analysis to enhance recognition accuracy, while leveraging advanced language processing to ensure meaningful and semantically rich outputs. By addressing challenges such as unstructured formats and distorted input quality, the framework facilitates seamless conversion of raw content into structured, machine-readable formats across domains like education, software development, and business process modeling.

The methodologies outlined in this work directly support core functionalities of diagram digitization systems, such as shape detection, text extraction from flowcharts, and structural relationship understanding. The emphasis on preprocessing and semantic validation strengthens the adaptability and reliability of automated recognition systems in handling diverse input sources including hand-drawn sketches, printed diagrams, and scanned documents.

2.2 Handwriting Recognition in Document Processing

This research focuses on applying deep learning techniques for automatic handwriting recognition in document analysis. The functionality of handwriting recognition systems includes image preprocessing, character segmentation, feature extraction, and sequence modeling using recurrent neural networks (RNNs) or transformer architectures. These systems aim to eliminate

manual transcription, reduce dependency on printed text, and enable digitization of handwritten notes, diagrams, and annotations.
The technologies used include convolutional neural networks (CNNs) for feature extraction from handwritten images, connectionist temporal classification (CTC) for sequence alignment, and attention-based mechanisms for improved accuracy. Despite significant progress, the literature identifies limitations such as difficulty recognizing varied handwriting styles, challenges with cursive text, sensitivity to image quality and lighting conditions, and the need for large, diverse training datasets. Additionally, preserving line breaks and formatting during text extraction remains challenging, particularly for diagram labels containing multiple lines of text.

2.3 AI-Powered Code and Flowchart Generation

This study explores the application of large language models (LLMs) for generating code and flowcharts from natural language descriptions. The functionality of these systems includes understanding user intent, breaking down algorithms into logical steps, mapping steps to appropriate flowchart symbols, and generating structured outputs in machine-readable formats. These tools aim to bridge the gap between conceptual algorithm design and visual representation, making programming more accessible to beginners and accelerating documentation for experienced developers.
The technologies discussed include transformer-based language models like GPT (Generative Pre-trained Transformer) and Gemini, prompt engineering techniques for guiding model outputs, and structured generation formats like JSON for representing flowchart components. However, the review highlights limitations such as inconsistency in generated outputs, difficulty handling complex nested logic, challenges in maintaining naming consistency across nodes, and the need for post-generation validation and editing. Furthermore, generating appropriate connection paths and maintaining proper flow direction remains challenging for current models.

2.4 Interactive Diagramming Tools and Libraries

This research examines various web-based diagramming libraries and their capabilities for creating interactive, editable diagrams. The functionality of these tools includes drag-drop interfaces for adding shapes, connection handling between nodes, zooming and panning for large diagrams, and export capabilities for sharing and integration. Libraries like React Flow, GoJS, and mxGraph provide developers with building blocks for creating custom diagramming applications tailored to specific domains.
The technologies used include SVG (Scalable Vector Graphics) for rendering, canvas-based drawing for performance, state management libraries for handling diagram data, and custom rendering for specialized node types. The literature identifies advantages such as rapid development, extensive customization options, and active community support. However, limitations include performance degradation with very large diagrams, complexity in implementing custom behaviors, varying support for different export formats, and challenges in achieving consistent appearance across different browsers.

2.5 Optical Graph Recognition Techniques

This study focuses specifically on optical graph recognition (OGR)—the task of converting images of graphs and diagrams into machine-readable representations. The functionality of OGR systems includes detecting nodes (shapes), recognizing node types (rectangle, diamond, circle, parallelogram), extracting text labels from nodes, identifying edges (connections) between nodes, and determining edge direction. These systems aim to automate the digitization of hand-drawn or printed graphs, flowcharts, and network diagrams.
The techniques employed include traditional computer vision methods like Hough transforms for line detection, contour analysis for shape identification, and graph-theoretic approaches for connection analysis. More recent approaches leverage deep learning for end-to-end diagram understanding. The literature identifies persistent challenges including handling overlapping elements, detecting arrows and direction indicators, distinguishing between different shape types with similar appearances, and accurately extracting text from within shapes. The combination of vision techniques with language models shows promise for addressing these challenges by incorporating contextual understanding.

2.6 Summary

The proposed Automated Diagram Digitization system aims to overcome the limitations identified in existing literature by providing:

- A vision-based recognition system that automatically identifies flowchart shapes, text, and connections from uploaded images using Google's Gemini AI.
- Support for multiple input methods including image upload, natural language descriptions for text-to-flowchart generation, and JSON import for existing flow data.
- An interactive editor with comprehensive features including drag-drop from shape palette, undo/redo with snapshot history, auto-layout for clean alignment, and edge labeling via double-click.
- Text customization options including bold, italic, and font size adjustments for node content to enhance readability.
- Multiple export formats including PNG for sharing, SVG for vector graphics, and JSON for data portability.
- Theme switching (dark/light mode) with persistent user preference for enhanced user experience.
- Persistent storage using PostgreSQL database for saving and retrieving flowcharts with UUID-based identification.

Unlike many existing systems that focus solely on recognition without editing capabilities or provide editing without automation, Automated Diagram Digitization combines AI-powered recognition with a feature-rich interactive editor. By integrating Google's Gemini AI for accurate shape and text extraction with React Flow for intuitive diagram manipulation, the system offers a complete workflow from hand-drawn sketch to editable digital diagram. The modular architecture and comprehensive feature set position this project as a practical, accessible solution for students, educators, software developers, and business professionals who need to digitize and edit flowcharts efficiently.

Thus, this project contributes to the growing field of intelligent document processing by offering an integrated, user-centered solution for automated diagram digitization that addresses the gaps identified in existing literature.

Chapter 3

System Study

The system study phase involves understanding existing diagram digitization challenges and analyzing requirements for the Automated Diagram Digitization system. This chapter covers problem definition, feasibility analysis, and requirements gathering. A comprehensive investigation was conducted through literature review, analysis of existing diagramming tools, and identification of user needs to establish a solid foundation for system design and implementation.

3.1 Existing System

3.1.1 Current Diagram Digitization Challenges

The existing process of digitizing flowcharts and diagrams faces several critical challenges that impact user productivity and diagram quality:

- **Manual Redrawing Required:** Users must manually recreate diagrams using tools like Microsoft Visio, Lucidchart, or draw.io, a time-consuming process that can take 15-30 minutes for moderately complex flowcharts. This not only wastes time but also introduces errors in the recreation process.
- **Lack of Automated Recognition:** Current systems cannot automatically identify shapes, text, and connections from images of hand-drawn or printed flowcharts. Users must manually identify each shape, type the text, and draw connections, defeating the purpose of digitization.
- **Text Extraction Challenges:** Standard OCR tools struggle with handwritten text and cannot preserve line breaks or formatting within shapes. When text spans multiple lines within a shape, the formatting is often lost, resulting in unreadable merged text.
- **Connection Detection Gap:** Existing systems fail to recognize the directional relationships between shapes. Even when shapes are detected, the arrows and connections that define the flow logic must be manually drawn, which is particularly problematic for complex flowcharts with multiple decision paths.
- **Limited Input Methods:** Most tools only support manual creation through drag-drop interfaces, lacking support for image upload, text-to-flowchart generation from algorithm descriptions, or JSON import for existing flow data.
- **No Intelligent Editing Features:** Automated tools typically lack post-recognition editing capabilities such as auto-layout for clean alignment, undo/redo for error correction, text styling for readability, and edge labeling for decision branches.

- **Export Limitations:** Many tools offer limited export options, restricting users to a single format that may not suit their needs for documentation, sharing, or further processing.

3.1.2 Disadvantages of Existing System

The limitations of current diagram digitization approaches can be summarized as follows:

- No automated shape recognition capability for hand-drawn or printed flowcharts
- Inability to detect connections and directional relationships between nodes
- Poor handling of handwritten text within diagram shapes
- Limited export options (typically only image formats, no SVG or JSON)
- No intelligent text extraction with line break preservation
- Lack of customizable styling options for nodes and edges
- No undo/redo functionality in most automated tools
- No support for text-to-flowchart generation from algorithm descriptions
- Absence of auto-layout for clean diagram alignment after recognition
- No theme switching (dark/light mode) for user comfort

3.2 Proposed System

3.2.1 System Overview

The Automated Diagram Digitization system is a web-based platform that combines AI-powered recognition with an interactive diagram editor. The system provides a complete workflow from hand-drawn sketch to editable digital diagram:

- **AI-Powered Recognition:** Uses Google's Gemini AI with custom prompts to analyze uploaded flowchart images, identify shapes (process, decision, start/end, input/output), extract embedded text, and recognize connection paths between nodes.
- **Multiple Input Methods:** Supports three input modes—image upload for existing diagrams, text description for algorithm-to-flowchart conversion, and JSON import for existing flow data.
- **Interactive Diagram Editor:** Provides drag-drop from shape palette, undo/redo with snapshot history, auto-layout for clean alignment, node resizing with text adaptation, and edge labeling via double-click.
- **Text Customization:** Enables text formatting including bold, italic, and font size adjustments for node content to enhance readability and presentation quality.

- **Export Functionality:** Supports multiple export formats including PNG for sharing, SVG for vector graphics, and JSON for data portability.
- **Theme Support:** Implements dark/light mode toggle with persistent user preference stored in localStorage for comfortable viewing.
- **Persistent Storage:** Enables saving and retrieving flowcharts using PostgreSQL database with UUID-based identification and timestamp tracking.

3.2.2 Advantages of Proposed System

- Automatic shape identification eliminates manual redrawing, saving significant time and effort
- Text extraction with proper line breaks preserves readability and original formatting
- Connection detection automatically creates edges with correct directional arrows
- Multiple input methods provide flexibility for different user workflows
- Comprehensive editing features match professional diagramming tools
- Export options support various use cases from documentation to data portability
- Undo/redo prevents accidental changes and enables experimentation
- Auto-layout ensures professional appearance with properly aligned diagrams
- Theme switching reduces eye strain during extended editing sessions
- Persistent storage enables workflow continuity across sessions

3.3 Feasibility Study

3.3.1 Technical Feasibility

The project utilizes proven, mature technologies that have demonstrated reliability across numerous production deployments:

- **Backend:** FastAPI provides high-performance asynchronous API endpoints with automatic OpenAPI documentation, making development and testing efficient.
- **AI Integration:** Google Gemini API offers state-of-the-art multimodal capabilities for image analysis and text generation, with extensive documentation and community support.
- **Frontend:** React 18 with Vite provides optimal performance with fast refresh and efficient rendering, while React Flow offers specialized diagramming capabilities.

- **Database:** PostgreSQL with JSONB support provides robust data persistence with ACID compliance and flexible schema for storing complex flowchart data.
- **Deployment:** The modular architecture supports cloud deployment with horizontal scaling capabilities as user numbers grow.

Each technology has extensive community support and a track record of handling production-scale applications, reducing implementation risk.

3.3.2 Operational Feasibility

- **User-Friendly Interface:** The input page features a tabbed interface with clear options for image upload, text prompt, and JSON import, requiring minimal user training.
- **Intuitive Editor:** Drag-drop from shape palette, familiar undo/redo shortcuts (Ctrl+Z, Ctrl+Y), and double-click edge labeling follow common software conventions.
- **Clear Feedback:** Visual indicators show upload progress, processing status, and error messages with actionable guidance.
- **Minimal Maintenance:** The system is designed for automatic error recovery with fallback mechanisms for malformed AI responses.
- **Responsive Design:** The interface adapts to different screen sizes, ensuring usability on desktop, tablet, and mobile devices.

3.3.3 Economic Feasibility

- **Open-Source Technologies:** FastAPI, React, PostgreSQL, and related libraries are open-source with no licensing costs, minimizing initial investment.
- **Cost-Effective AI:** Google Gemini API offers competitive pricing with free tier options for development and testing phases.
- **Reduced Labor Costs:** Automation of diagram digitization eliminates hours of manual redrawing, providing significant time savings for users.
- **Scalable Architecture:** The system can be deployed on cloud infrastructure with pay-as-you-grow pricing, avoiding large upfront infrastructure costs.
- **Return on Investment:** Educational institutions and businesses can achieve positive ROI through improved productivity and reduced manual effort.

3.4 Requirements Analysis

3.4.1 Functional Requirements

Based on user needs and system objectives, the following functional requirements have been identified:

3.4.1.1 Input Module Requirements

- **Image Upload:** Support PNG, JPG, and WebP image formats with drag-drop interface, file size validation, and preview functionality.
- **Text Prompt:** Provide multi-line textarea for algorithm descriptions with character count and example placeholder.
- **JSON Import:** Accept JSON file upload with validation and error messaging for malformed data.

3.4.1.2 AI Processing Requirements

- **Shape Recognition:** Identify four basic shape types—process (rectangle), decision (diamond), start/end (oval), and input/output (parallelogram)—from uploaded images.
- **Text Extraction:** Extract text from shapes with proper line break preservation using \n characters for multi-line labels.
- **Connection Detection:** Detect edges between nodes with directional arrows and assign appropriate handles (top, bottom, left, right).
- **Coordinate Estimation:** Calculate normalized coordinates on a 0-1000 scale for consistent positioning across different image sizes.
- **Error Handling:** Parse AI responses reliably with regex fallback for extracting JSON from text responses.

3.4.1.3 Editor Module Requirements

- **Drag-Drop Functionality:** Allow users to drag shapes from sidebar palette and drop onto canvas at desired positions.
- **Node Resizing:** Support node resizing with minimum dimensions and text adaptation to fit resized nodes.
- **Undo/Redo:** Implement snapshot-based history with keyboard shortcuts (Ctrl+Z for undo, Ctrl+Y or Ctrl+Shift+Z for redo).
- **Auto-Layout:** Provide one-click auto-layout using column-based clustering algorithm for clean alignment.
- **Edge Labeling:** Enable double-click on edges to add and edit labels with proper background styling.
- **Deletion:** Support deletion of nodes and edges using Delete or Backspace keys with snapshot capture.

- **Color Customization:** Allow users to change node and edge colors through color picker interface.
- **Text Styling:** Provide controls for bold, italic, and font size adjustments for selected nodes.

3.4.1.4 Export Module Requirements

- **PNG Export:** Capture the entire canvas as PNG image using html-to-image library.
- **SVG Export:** Generate SVG vector graphics using React Flow's SVG renderer.
- **JSON Export:** Serialize current nodes and edges to JSON format with proper indentation.

3.4.1.5 Storage Module Requirements

- **Save Flowchart:** Store flowchart data in PostgreSQL database with UUID primary key and timestamp tracking.
- **Retrieve Flowchart:** Fetch flowchart by UUID and reconstruct editor state.
- **Update Flowchart:** Modify existing flowchart title or flow data with updated_at timestamp refresh.

3.4.2 Non-Functional Requirements

3.4.2.1 Performance Requirements

- **Image Analysis:** Complete image analysis within 3-5 seconds for standard flowchart images (up to 1024x1024 pixels).
- **Text-to-Flowchart:** Generate flowchart from text prompt within 2-3 seconds.
- **Editor Operations:** Execute node drag-drop, undo/redo, and auto-layout within 100ms for smooth user experience.
- **API Response:** Maintain API response times under 200ms for standard database operations.
- **Page Load:** Achieve initial page load under 2 seconds with optimized asset delivery.
- **Concurrent Users:** Support at least 50 concurrent users without performance degradation.

3.4.2.2 Usability Requirements

- **Intuitive Interface:** Design interface requiring minimal training for users with basic computer skills.
- **Clear Error Messages:** Provide actionable error messages with guidance for resolution.
- **Responsive Design:** Ensure comfortable usage on desktop monitors, tablets, and mobile phones.
- **Theme Consistency:** Maintain consistent appearance across dark and light themes with smooth transitions.
- **Helpful Tooltips:** Include tooltips for all toolbar buttons explaining their functionality.
- **Keyboard Shortcuts:** Support common keyboard shortcuts for undo/redo and delete operations.

3.4.2.3 Reliability Requirements

- **Uptime:** Maintain 99% availability during normal business hours.
- **Error Recovery:** Implement graceful error handling with fallback mechanisms for AI processing failures.
- **Data Persistence:** Ensure data integrity with transaction support for database operations.
- **Input Validation:** Validate all user inputs on both client and server sides to prevent invalid data.
- **Backup:** Support database backup and restore procedures for disaster recovery.

3.4.2.4 Security Requirements

- **API Key Protection:** Store Gemini API key in environment variables, never in client-side code.
- **Input Sanitization:** Sanitize all user inputs to prevent injection attacks.
- **CORS Configuration:** Restrict API access to allowed origins (localhost for development, production domain for deployment).
- **SQL Injection Prevention:** Use parameterized queries for all database operations.
- **Environment Separation:** Maintain separate configuration for development, testing, and production environments.

Three-Tier Architecture

- **Presentation Layer:** React 18, React Flow, Tailwind CSS
- **Application Layer:** FastAPI (Python), Gemini AI Integration
- **Data Layer:** PostgreSQL Database with JSONB Support

Figure 3.1: Automated Diagram Digitization Three-Tier Architecture

3.5 System Architecture Overview

The Automated Diagram Digitization system follows a three-tier architecture with clear separation of concerns:

- **Presentation Layer:** Handles all user interaction through React components, including the input page with tabs for different input methods, the diagram editor with React Flow, and the sidebar with draggable shape palette.
- **Application Layer:** Implements business logic using FastAPI, exposing RESTful endpoints for image analysis, text-to-flowchart generation, and database operations. Integrates with Google Gemini AI for intelligent recognition.
- **Data Layer:** Manages persistence through PostgreSQL database with JSONB column for storing complete React Flow state, enabling flexible schema while maintaining query performance.

This separation enables each tier to evolve independently, with the ability to scale components based on demand.

3.6 System Modules

3.6.1 Input Module

Handles three input methods: image upload with drag-drop interface, text prompt with multiline textarea, and JSON import with file upload. Validates input formats, provides visual feedback, and prepares data for AI processing.

3.6.2 AI Processing Module

Interfaces with Google Gemini AI using custom system prompts for image analysis and text-to-flowchart conversion. Parses JSON responses, handles errors with regex fallback, and ensures proper ID generation for edges.

3.6.3 Editor Module

Renders nodes and edges using React Flow with custom node components for each shape type. Implements drag-drop from sidebar, undo/redo with snapshot history, auto-layout with column-based clustering, edge labeling via double-click, and comprehensive styling options.

3.6.4 Export Module

Provides PNG export using html-to-image library, SVG export using React Flow's SVG renderer, and JSON export of current flow state. Handles file download with appropriate naming based on flowchart title.

3.6.5 Storage Module

Manages PostgreSQL database operations including save, retrieve, and update of flowcharts. Implements UUID-based identification, timestamp tracking, and JSONB storage for flow data.

3.7 User Roles and Permissions

Role	Permissions
User (All)	Upload images, enter text prompts, import JSON
User (All)	Create, edit, and delete flowcharts
User (All)	Export diagrams (PNG, SVG, JSON)
User (All)	Toggle theme (dark/light mode)
User (All)	Save and retrieve flowcharts from database

Table 3.1: User Roles and Access Levels

The system currently implements a single user role with full access to all features, as authentication is not required for basic usage. Future enhancements may include user authentication with role-based access for collaborative features.

3.8 Summary

The system study has established clear requirements for the Automated Diagram Digitization project, addressing critical gaps in automated shape recognition, text extraction, connection detection, and interactive editing. The analysis of existing systems revealed significant inefficiencies in manual redrawing, lack of automation, and limited editing capabilities.

The proposed system directly addresses these challenges through AI-powered recognition, multiple input methods, comprehensive editing features, and flexible export options. Feasibility analysis confirms that the solution is technically achievable using proven technologies (FastAPI, React, PostgreSQL, Gemini AI), operationally practical with an intuitive user interface, and economically beneficial through time savings and open-source cost structure.

The modular architecture with clear separation of concerns positions Automated Diagram Digitization to effectively transform how users convert hand-drawn diagrams into editable digital formats, improving productivity and workflow efficiency for students, educators, software developers, and business professionals.

Chapter 4

System Design

This chapter presents the comprehensive system design of the Automated Diagram Digitization platform, detailing the architectural decisions, data flow, database schema, API design, and module specifications. The system design follows industry best practices for web applications, ensuring scalability, maintainability, security, and seamless integration between the frontend editor, backend services, and AI processing engine.

4.1 System Architecture Overview

The Automated Diagram Digitization platform is built on a robust three-tier architecture ensuring scalability, maintainability, and security while providing seamless integration between the React frontend, FastAPI backend, and PostgreSQL database, with clear separation of concerns between presentation, business logic, and data persistence layers.

Figure 4.1: Three-Tier System Architecture of Automated Diagram Digitization Platform

4.1.1 Presentation Layer (Client Tier)

The presentation layer serves as the user interface through which all users interact with the platform. It utilizes React 18 with Vite for fast development and optimal performance, React Flow for interactive diagram rendering, and Tailwind CSS for responsive, modern design. Key advantages include component-based architecture for reusability, fast refresh during development, and a consistent experience across all devices and screen sizes.

The interface is organized into two main sections: the input page with three tabs for image upload, text prompt, and JSON import; and the flowchart editor with a draggable shape sidebar, canvas area with background grid, and toolbar with undo/redo, auto-layout, export, and theme toggle controls.

4.1.2 Application Layer (Business Tier)

The application layer implements all business logic, validation, workflow orchestration, and AI integration using FastAPI with Python. Key components include RESTful API endpoints for image analysis and text-to-flowchart generation, integration with Google Gemini AI using custom prompts, request validation using Pydantic models, centralized error handling with regex fallback for malformed responses, and asynchronous database operations using SQLAlchemy.

4.1.3 Data Layer (Database Tier)

The data layer manages all persistent storage using PostgreSQL with JSONB support for flexible schema storage. Key features include UUID v4 primary keys for distributed generation without coordination, JSONB columns for storing complete React Flow state (nodes and edges), timestamp tracking with timezone support for created_at and updated_at fields, foreign key constraints where applicable, and connection pooling for optimal performance.

4.2 Data Flow Diagrams

Data Flow Diagrams (DFDs) provide a graphical representation of how information moves through the Automated Diagram Digitization platform. This section describes the three most critical workflows.

4.2.1 Image Upload and Analysis Flow

The image analysis process enables users to upload flowchart images for automatic digitization. The workflow proceeds as follows: the user selects an image file via drag-drop or file picker, the frontend validates file type and size, the image is sent to the backend via multipart/form-data, the backend processes the image using Pillow (resizing to thumbnail), the image is sent to Gemini AI with the system prompt, the AI returns JSON with nodes and edges, the backend parses the response and ensures edge IDs exist, and the frontend renders the flowchart in the editor. The user can then edit, save, or export the diagram.

4.2.2 Text-to-Flowchart Generation Flow

The text-to-flowchart feature enables users to describe algorithms in natural language for automatic flowchart generation. The workflow proceeds as follows: the user enters a text description in the prompt textarea, the frontend sends the prompt to the backend via POST request, the backend combines the user prompt with the text-to-flowchart system prompt, the combined

Figure 4.2: Image Upload and Analysis Data Flow Diagram

prompt is sent to Gemini AI, the AI returns JSON with nodes and edges (may include explanatory text), the backend extracts JSON using regex pattern matching, the parsed JSON is returned to the frontend, and the editor renders the generated flowchart. The user can then modify the generated diagram as needed.

4.2.3 JSON Import and Export Flow

The JSON import/export functionality enables data portability and integration with other tools. For import: the user uploads a JSON file or pastes JSON into textarea, the frontend validates JSON format, the validated JSON is passed directly to the editor for rendering. For export: the user clicks the export button and selects format (JSON/PNG/SVG), for JSON export the current nodes and edges are serialized to JSON and downloaded as a file, for PNG export the canvas is captured using html-to-image and downloaded, for SVG export the React Flow SVG renderer generates vector graphics for download.

4.3 Database Schema Design (PostgreSQL)

The Automated Diagram Digitization platform utilizes PostgreSQL with JSONB support, providing reliable data persistence, ACID compliance, and flexible schema for storing complex flowchart data. The schema uses UUID primary keys, timestamp tracking, and JSONB columns for flow data storage.

The Entity-Relationship diagram above illustrates the database structure. At the center is the `flowcharts` table, which stores all flowchart data. Each flowchart has a unique UUID identifier, a title string, a JSONB column containing the complete React Flow state (nodes and edges), and timestamp columns for creation and last update tracking.

Figure 4.3: Database Entity-Relationship Diagram

The JSONB column provides flexibility to store varying node and edge structures without requiring schema changes. Each node object contains id, type, position coordinates, data (including label), width, height, and optional styling properties. Each edge object contains id, source, target, optional label, type (smoothstep/straight), markerEnd for arrows, and optional styling properties.

The schema follows best practices for web applications, uses `created_at` and `updated_at` timestamps with timezone support for audit compliance, applies indexes on frequently queried columns, and uses UUID primary keys for security benefits over sequential integers. The JSONB column supports indexing on specific fields if needed for performance.

All timestamp columns use `TIMESTAMPTZ` rather than a timezone-naive type, storing values in UTC and allowing the application layer to convert to the user's local timezone at render time. This is particularly important for a platform that may be deployed across different geographical regions. Consistent UTC storage eliminates ambiguity in timestamp comparisons.

The schema is managed through a versioned migration system using Alembic (SQLAlchemy migrations), capturing every structural change as an incremental script across development, staging, and production environments. This ensures consistent database evolution across all deployments.

The following table provides a concise summary of the `flowcharts` table structure as implemented in PostgreSQL.

Column	Data Type	Description & Constraints
id	UUID	Primary key, default gen_random_uuid()
title	VARCHAR(255)	Flowchart title, default 'Untitled'
flow_data	JSONB	Complete node/edge data, NOT NULL
created_at	TIMESTAMPTZ	Creation timestamp, default now()
updated_at	TIMESTAMPTZ	Last update timestamp, default now()

Table 4.1: Flowcharts Table Schema

Database Indexes for Performance Optimization:

- `CREATE INDEX idx_flowcharts_id ON flowcharts(id);` – Primary key index for fast lookup by UUID.
- `CREATE INDEX idx_flowcharts_created_at ON flowcharts(created_at);` – Optimizes queries sorting by creation date.
- `CREATE INDEX idx_flowcharts_updated_at ON flowcharts(updated_at);` – Optimizes queries sorting by update date.
- `CREATE INDEX idx_flowcharts_title ON flowcharts(title);` – Enables fast search by title (for future search functionality).
- `CREATE INDEX idx_flowcharts_gin_nodes ON flowcharts USING gin ((flow_data -> 'nodes'));` – GIN index for querying inside JSONB nodes array.

Sample flow_data JSON Structure:

```
{
  "nodes": [
    {
      "id": "node_123",
      "type": "process",
      "position": { "x": 100, "y": 200 },
      "data": { "label": "Process Data" },
      "width": 150,
      "height": 80,
      "style": {
        "backgroundColor": "#ffffff",
```

```
        "color": "#000000",
        "fontSize": 12,
        "fontWeight": "normal",
        "fontStyle": "normal"
      }
    },
    {
      "id": "node_456",
      "type": "decision",
      "position": { "x": 300, "y": 200 },
      "data": { "label": "Is Valid?\\nYes/No" },
      "width": 120,
      "height": 120
    }
  ],
  "edges": [
    {
      "id": "e1-2",
      "source": "node_123",
      "target": "node_456",
      "sourceHandle": "bottom",
      "targetHandle": "top",
      "label": "Yes",
      "type": "smoothstep",
      "markerEnd": { "type": "arrowclosed" },
      "style": { "stroke": "#000000" },
      "labelStyle": {
        "fill": "#000000",
        "fontWeight": 600
      },
      "labelBgStyle": {
        "fill": "#ffffff",
        "stroke": "#000000",
        "strokeWidth": 1,
        "rx": 4,
        "ry": 4
      }
    }
  ]
}
```

Listing 4.1: Sample flow_data JSON Structure

4.4 API Design (RESTful Endpoints)

The Automated Diagram Digitization platform implements a comprehensive RESTful API following industry best practices, serving as the communication backbone between the React frontend and FastAPI backend. Endpoints are resource-oriented, use appropriate HTTP methods, and consistent JSON response formats.
The API is organised into logical resource groups: image analysis, text-to-flowchart generation, and flowchart management (CRUD operations). Middleware for CORS handling, request validation, and error handling is applied consistently. Incoming payloads are validated for type correctness, required fields, and domain constraints before any business logic executes.
Error responses follow a consistent JSON structure containing detail message and appropriate HTTP status codes: `200` for successful retrieval, `201` for resource creation, `400` for validation failures, `404` when a requested resource does not exist, and `500` for internal server errors.

Method	Endpoint	Description	Request Body
POST	/analyze	Upload flowchart image for AI analysis	multipart/form-data (image file)
POST	/analyze/algorithm	Generate flowchart from text description	JSON: { ”prompt”: ”text” }
POST	/flowcharts	Create new flowchart in database	JSON: { ”title”: ”...”, ”flow_data”: {...} }
GET	/flowcharts/{id}	Retrieve flowchart by UUID	-
PUT	/flowcharts/{id}	Update existing flowchart	JSON with partial updates
GET	/	Health check endpoint	-

Table 4.2: Core API Endpoints

Sample Request - Image Analysis:

```
POST /analyze HTTP/1.1
Host: localhost:8000
Content-Type: multipart/form-data; boundary=----WebKitFormBoundary

------WebKitFormBoundary
Content-Disposition: form-data; name="file"; filename="flowchart.png"
Content-Type: image/png

[image data]
```

```
------WebKitFormBoundary--
```

Listing 4.2: Image Analysis API Request

Sample Response - Image Analysis:

```
{
  "nodes": [
    {
      "id": "1",
      "type": "start",
      "position": { "x": 500, "y": 50 },
      "data": { "label": "Start" },
      "width": 90,
      "height": 40
    },
    {
      "id": "2",
      "type": "process",
      "position": { "x": 475, "y": 150 },
      "data": { "label": "Enter\\nCredentials" },
      "width": 150,
      "height": 80
    }
  ],
  "edges": [
    {
      "id": "e1-2",
      "source": "1",
      "target": "2",
      "sourceHandle": "bottom",
      "targetHandle": "top",
      "type": "smoothstep",
      "markerEnd": { "type": "arrowclosed" }
    }
  ]
}
```

Listing 4.3: Image Analysis API Response

4.5 Module-wise Detailed Design

This section provides design specifications for each major module of the Automated Diagram Digitization platform, describing their component structure and interactions with other modules.

4.5.1 Input Module

The input module handles all user input methods before AI processing. Components include:

- **Image Upload Component**: Drag-drop zone with file picker fallback, file type validation (PNG, JPG, WebP), file size validation, preview thumbnail, and upload progress indicator.
- **Text Prompt Component**: Multi-line textarea with placeholder text, character count (optional), example prompt button, and submit with loading state.
- **JSON Import Component**: Textarea for JSON paste with syntax highlighting, file upload button for .json files, JSON validation with error messaging, and format preview option.
- **Tab Navigation**: Tabbed interface switching between input methods with active state styling and form reset on tab change.

4.5.2 AI Processing Module

The AI processing module interfaces with Google Gemini AI and handles response parsing. Components include:

- **Gemini API Client**: Configured with API key from environment variables, handles authentication, and sends requests with appropriate system prompts.
- **Image Processor**: Uses Pillow to open uploaded images, resizes to thumbnail (1024x1024 max) for optimal API performance, and maintains aspect ratio.
- **Prompt Manager**: Stores and manages system prompts for image analysis and text-to-flowchart generation, combines user input with system prompts.
- **Response Parser**: Parses JSON responses from Gemini, handles malformed responses using regex extraction (\(\.*\)), ensures edge IDs exist (generates e{source}-{target} if missing), and validates required fields.
- **Error Handler**: Catches and logs API errors, returns user-friendly error messages, and provides fallback responses when possible.

4.5.3 Editor Module

The editor module is the core interactive component for diagram creation and modification. Components include:

- **React Flow Canvas**: Renders nodes and edges with background grid, supports zooming and panning, and handles selection and deletion.

- **Sidebar Palette**: Displays draggable shape icons (start, process, decision, io) with SVG icons, implements HTML5 drag-drop with data transfer, and provides visual feedback during drag.
- **Custom Node Components**: BaseNode with common functionality, ProcessNode with rectangle shape, StartNode with rounded rectangle (pill shape), DecisionNode with diamond shape (clip-path polygon), and InputNode with parallelogram shape (skew transform).
- **Node Resizer**: Uses React Flow's NodeResizer component, visible only when node selected, maintains minimum dimensions (80x40 for most nodes, 120x120 for decision), and preserves text readability.
- **Edge Label Editor**: Double-click handler on edges, inline input popup at mouse position, Enter key saves, Escape cancels, and blur saves automatically.
- **Undo/Redo System**: Snapshot-based history using useState, takeSnapshot before operations, undo restores previous state, redo restores next state, and keyboard shortcuts (Ctrl+Z, Ctrl+Y, Ctrl+Shift+Z).
- **Auto-Layout Engine**: Column-based clustering algorithm, calculates true center X for each node, clusters within tolerance (150px), snaps nodes to average column positions, and updates edge types based on alignment.
- **Styling Controls**: Color picker for nodes and edges, text styling buttons (bold, italic), font size dropdown (10-20px), and style application to selected elements.

4.5.4 Export Module

The export module handles all diagram export functionality. Components include:

- **PNG Exporter**: Uses html-to-image library to capture canvas element, creates download link with data URL, and names file using flowchart title.
- **SVG Exporter**: Leverages React Flow's built-in SVG export capabilities, generates vector graphics with proper scaling, and maintains styling.
- **JSON Exporter**: Serializes current nodes and edges to JSON string with pretty printing (2-space indent), creates blob with application/json type, and triggers download.
- **Export Menu**: Dropdown menu with export options, icons for each format, and disabled state during export operations.

4.5.5 Storage Module

The storage module manages database operations for flowchart persistence. Components include:

- **Database Client**: Async SQLAlchemy engine with connection pooling, configured with DATABASE_URL from environment, and SSL enabled for production.
- **Create Operation**: Inserts new flowchart with title and flow_data, returns generated UUID and timestamps, and handles validation errors.
- **Read Operation**: Retrieves flowchart by UUID using SELECT query, returns 404 if not found, and parses JSONB flow_data.
- **Update Operation**: Updates title or flow_data with partial updates, refreshes updated_at timestamp, and returns updated record.

4.5.6 Theme Module

The theme module manages dark/light mode switching. Components include:

- **Theme Context**: React context providing theme state and toggle function, consumed by all components needing theme awareness.
- **Theme Toggle Button**: Button with sun/moon icons, toggles between dark and light modes, and updates localStorage with preference.
- **CSS Variables**: Root-level CSS variables for node backgrounds, borders, text colors, and canvas background, dynamically updated on theme change.
- **Persistence**: localStorage stores user theme preference, initial theme loaded from localStorage or defaults to dark mode.

4.6 AI Prompt Design

The system uses two specialized prompts for Gemini AI, carefully engineered for accurate flowchart recognition and generation.

4.6.1 Image Analysis System Prompt

```
SYSTEM_PROMPT = """
You are an expert Optical Graph Recognition engine.
Analyze the provided flowchart image and extract its structure into a JSON
    format compatible with React Flow.

Follow these strict rules:
```

```

1. Nodes:
- Identify shapes and map to these EXACT types:
  - rectangle -> type: "process"
  - diamond -> type: "decision"
  - oval/circle -> type: "start"
  - parallelogram -> type: "io"
- Dimensions: Calculate the actual width and height of each shape based on its bounding box in the image (relative to a 1000x1000 canvas).
- Extract text into data.label (use \\n for line breaks).
- Estimate coordinates on a 0-1000 scale.

2. Edges:
- 'type': "smoothstep"
- 'markerEnd': { "type": "arrowclosed" }
- 'sourceHandle' and 'targetHandle': Use "top", "bottom", "left", or "right" based on where the arrow physically enters/exits the shape.

3. Output format:
- Return ONLY raw JSON. No markdown, no backticks, no preamble.
- Structure: {"nodes": [...], "edges": [...]}
"""
```

Listing 4.4: Image Analysis System Prompt

4.6.2 Text-to-Flowchart System Prompt

```
TEXT_TO_FLOW_PROMPT = """
You are a React Flow architect. Convert the user's process description or algorithm into a structured JSON.
Output ONLY raw JSON.

NODE RULES:
- TYPES: Use only: "start", "process", "decision", "io".
- Include width and height at the root level.
- Scale nodes based on text length (increase width/height for multi-line labels).
- For decision nodes with long text, increase size to at least 200x200.

EDGE RULES:
- Use "type": "smoothstep" and "markerEnd": {"type": "arrowclosed"}.
- Use "sourceHandle" and "targetHandle" (top, bottom, left, right) for clean connections.
- For decision branches (Yes/No), include label styling.

```

```
LABEL RULES:
- Use \\n for line breaks.
- For decision nodes, format text to fit within diamond shape.
- Include labelStyle for edge labels with fill, fontWeight, and background
    styling.
"""
```

Listing 4.5: Text-to-Flowchart System Prompt

4.7 Algorithm Design

The Automated Diagram Digitization platform incorporates several internal algorithms to support diagram generation, layout organization, and editing functionality. These algorithms ensure that generated flowcharts remain visually structured, readable, and easy to modify. Two important algorithms implemented in the system are the column-based auto layout algorithm and the snapshot-based undo and redo algorithm.

4.7.1 Column-based Auto Layout Algorithm

The column-based auto layout algorithm is responsible for organizing nodes in a flowchart so that they appear visually aligned and structured. When diagrams are generated automatically by the AI system or manually edited by users, node positions may not always be perfectly aligned. The auto layout algorithm corrects this by grouping nodes into vertical columns based on their horizontal positions.

The algorithm begins by determining the maximum width required for each node type. Node types such as start nodes, process nodes, decision nodes, and input/output nodes may have different sizes. By determining the largest width for each type, the algorithm ensures visual consistency across the diagram.

After establishing the node widths, the algorithm calculates the horizontal center position of each node using its current position and width. These center positions are then used to sort the nodes along the horizontal axis. Sorting enables the algorithm to identify nodes that are positioned near one another.

Once sorted, the algorithm performs a clustering process in which nodes whose center positions fall within a defined tolerance range are grouped together into a single column. Each column maintains an average center value calculated from all nodes assigned to that group.

In the final stage, the algorithm repositions each node so that its center aligns with the calculated average center of its column. This ensures that nodes belonging to the same logical stage of a flowchart appear vertically aligned while maintaining their original vertical ordering. As a result, the flowchart becomes visually balanced and easier to interpret.

4.7.2 Undo and Redo Snapshot Algorithm

The platform includes an undo and redo mechanism that allows users to revert or restore modifications made while editing diagrams. This feature is implemented using a snapshot-based history management algorithm.

Whenever a user performs an operation that modifies the diagram, such as adding a node, deleting an edge, moving an element, or changing text content, the system captures a snapshot representing the complete state of the diagram at that moment. Each snapshot contains the full set of nodes and edges currently present in the flowchart.

Snapshots are stored in a history structure consisting of two stacks referred to as the *past* stack and the *future* stack. The past stack records previous states of the diagram, while the future stack stores states that can be restored through redo operations.

When a user performs an undo operation, the system retrieves the most recent snapshot from the past stack and restores the diagram to that earlier state. At the same time, the current state of the diagram is stored in the future stack, allowing it to be restored later if necessary.

If the user performs a redo operation, the system retrieves the next snapshot from the future stack and applies it to the diagram. The previously restored state is then returned to the past stack. This two-stack mechanism allows users to move backward and forward through their editing history without losing information.

Keyboard shortcuts such as `Ctrl+Z` for undo and `Ctrl+Y` or `Ctrl+Shift+Z` for redo provide quick access to these operations, improving usability during diagram editing.

Together, these algorithms ensure that diagrams remain organized and editable while providing users with flexible control over the editing process. The auto layout algorithm enhances visual clarity by maintaining alignment among nodes, while the snapshot-based history system provides reliable recovery from editing mistakes.

4.8 Error Handling Design

The system implements comprehensive error handling at multiple levels to ensure robustness.

4.8.1 AI Response Error Handling

- **Malformed JSON**: Uses regex extraction to find JSON block within text response: `re.search(r'(\{\.*\})', text, re.DOTALL)`.
- **Unescaped Characters**: Uses `json.loads(json_str, strict=False)` to handle unescaped newlines within strings.
- **Missing Fields**: Validates required fields (nodes, edges) and provides defaults if missing.
- **Missing Edge IDs**: Automatically generates IDs in format `e{source}-{target}` if not provided by AI.

4.8.2 API Error Handling

- **400 Bad Request**: Returned for missing prompt field, invalid JSON format, or validation failures.
- **404 Not Found**: Returned when flowchart ID does not exist in database.
- **500 Internal Server Error**: Returned for AI API failures, database errors, or unexpected exceptions with error logging.

4.8.3 Frontend Error Handling

- **File Upload Errors**: Validates file type and size before upload, displays user-friendly error messages.
- **JSON Import Errors**: Validates JSON syntax, displays line and column of error with descriptive message.
- **API Request Errors**: Catches fetch errors, displays toast notifications, and provides retry options.

4.9 Performance Optimization

The Automated Diagram Digitization platform implements multiple optimization strategies to ensure responsive user experiences.

Backend Optimization:

- **Image Resizing**: Thumbnail resizing to 1024x1024 max before sending to Gemini API reduces payload size and processing time.
- **Connection Pooling**: SQLAlchemy connection pooling with configurable pool size (5-20 connections) minimizes connection overhead.
- **Async Operations**: FastAPI async endpoints with await for database and AI operations improve concurrency.
- **JSONB Indexing**: GIN indexes on flow_data for potential future querying requirements.

Frontend Optimization:

- **React Flow Performance**: Uses virtualization for large diagrams, limits re-renders with memo and useCallback.
- **Debounced Operations**: Auto-save and auto-layout debounced to prevent excessive operations during rapid edits.
- **Lazy Loading**: Code splitting with React.lazy for editor component (loaded only when needed).

- **CSS Variables**: Theme switching uses CSS variable updates instead of full re-renders.

AI Optimization:

- **Prompt Engineering**: Temperature=0 for deterministic outputs, response_mime_type="application/json" for structured responses.
- **Image Optimization**: Thumbnail resizing reduces API latency and costs.
- **Caching**: Consider caching frequent prompts or responses for future implementation.

4.10 Summary

This chapter has presented the complete system design of the Automated Diagram Digitization platform, spanning the three-tier architecture, data flow diagrams, database schema, API design, module specifications, AI prompt design, algorithms, error handling, and performance optimization strategies. The design decisions reflect consistent priorities: accurate AI-powered recognition, intuitive user experience, robust error handling, and scalable architecture.

The layered separation between presentation, business logic, and data persistence ensures each tier can evolve independently. The database schema with UUID primary keys and JSONB support provides a flexible foundation for storing complex flowchart data. The RESTful API exposes this functionality in a predictable manner, while the AI integration with carefully engineered prompts ensures accurate shape recognition and text extraction.

The modular design with clear separation of concerns (input module, AI processing, editor, export, storage, theme) enables independent development and testing of each component. The auto-layout algorithm ensures professional diagram appearance, while the undo/redo system provides error recovery. The following chapter details the implementation of each module described here, translating these design specifications into working code.

Chapter 5

Implementation

This chapter details the implementation of the Automated Diagram Digitization system, describing each module's logic, API interactions, and integration with the broader application. The implementation follows a layered architecture: FastAPI handles HTTP routing, PostgreSQL serves as the persistent store, and React with Vite presents the user interface. All code is organised within separate frontend and backend directories, with environment-specific settings separated into a `.env` file.

5.1 Development Environment Setup

The backend runs on Python 3.9+ with the FastAPI framework. The database is a PostgreSQL instance with JSONB support for flexible schema storage. The frontend runs on Node.js v18+ with React 18 and Vite for fast development. Sensitive configuration values — database connection strings and Gemini API keys — are stored in a `.env` file at the project root and loaded via the `python-dotenv` package.

Figure 5.1: Project structure showing backend and frontend directories.

5.2 Backend Implementation

5.2.1 Image Analysis Endpoint

The image analysis endpoint accepts an uploaded flowchart image, processes it, sends it to Gemini AI with a custom system prompt, and returns structured JSON containing nodes and

edges. The image is resized to a thumbnail of 1024x1024 pixels for optimal API performance while maintaining aspect ratio.

Figure 5.2: Image upload interface with drag-drop area and file selection.

5.2.2 Text-to-Flowchart Endpoint

The text-to-flowchart endpoint accepts a natural language description of an algorithm and returns a generated flowchart. The user prompt is combined with a specialized system prompt designed for flowchart generation. The response is cleaned using regex extraction to handle any explanatory text the AI might add, ensuring only valid JSON is returned.

5.2.3 Database Operations

The database layer uses SQLAlchemy with async support for PostgreSQL operations. The flowcharts table stores each diagram with a UUID primary key, title, JSONB column for the complete React Flow state, and timestamps for creation and updates. The JSONB column provides flexibility to store varying node and edge structures without requiring schema changes.

5.3 Frontend Implementation

5.3.1 Input Page

The input page provides three methods for creating flowcharts: image upload with drag-drop support, text prompt for algorithm description, and JSON import for existing flow data. A tabbed interface allows users to switch between input methods, with each tab showing the appropriate form controls.

Figure 5.3: Input page showing three tabs for image upload, text prompt, and JSON import.

5.3.2 Sidebar with Draggable Shapes

The sidebar displays four shape types: start/end (oval), process (rectangle), decision (diamond), and input/output (parallelogram). Each shape is represented by an SVG icon and a label. HTML5 drag-drop is implemented, with the shape type stored in the dataTransfer object for retrieval when dropped on the canvas.

Figure 5.4: Sidebar with draggable shape icons.

5.3.3 Flowchart Editor

The flowchart editor is the core component for diagram creation and editing. It uses React Flow to render nodes and edges, with custom node components for each shape type. The canvas features a dot grid background, zoom and pan controls, and selection handling. Nodes can be resized using resize handles, with minimum dimensions enforced to maintain text readability.

5.3.4 Custom Node Components

Each shape type has its own node component that extends a base node. The base node provides common functionality: connection handles on all four sides, double-click editing, and node

Figure 5.5: Flowchart editor canvas with nodes, edges, and toolbar.

resizing. Shape-specific styling is applied using CSS: rectangles for process nodes, diamonds using clip-path polygons, rounded rectangles for start nodes, and skewed parallelograms for input/output nodes.

5.3.5 Undo/Redo Functionality

Undo/redo is implemented using a snapshot-based history system. Before any operation that modifies the diagram (node drag, edge creation, deletion, etc.), a snapshot of the current nodes and edges is saved to a past stack. Undo restores the previous state from the past stack and pushes the current state to a future stack. Keyboard shortcuts Ctrl+Z (undo), Ctrl+Y (redo), and Ctrl+Shift+Z (redo) are supported.

Figure 5.6: Toolbar showing undo, redo, and auto-layout buttons.

5.3.6 Auto-Layout Feature

The auto-layout feature uses a column-based clustering algorithm to align nodes neatly. It calculates the true center X-coordinate for each node, clusters nodes within a tolerance of 150 pixels, computes the average center for each cluster, and repositions nodes to align with their cluster's average. Edge types are automatically adjusted based on alignment — straight edges for vertically aligned nodes, smoothstep edges for others.

Figure 5.7: Auto-layout applied showing nodes aligned in columns.

5.3.7 Edge Label Editing

Edge labels can be added or edited by double-clicking on any edge. An inline input popup appears at the mouse position, allowing the user to enter label text. Pressing Enter saves the label, Escape cancels, and clicking elsewhere also saves. Labels are displayed with a background box for readability, with styling that adapts to the current theme.

5.3.8 Theme Implementation

The system supports both dark and light modes using CSS variables. A theme context provides the current theme state and toggle function. User preference is persisted in localStorage. CSS variables control node backgrounds, borders, text colors, canvas background, and grid colors, with all components reacting to theme changes.

5.3.9 Export Functionality

Three export formats are supported: PNG for sharing, SVG for vector graphics, and JSON for data portability. PNG export uses the html-to-image library to capture the canvas element. SVG export leverages React Flow's built-in SVG rendering. JSON export serializes the current nodes and edges with pretty printing.

Figure 5.8: Edge label editing popup on a decision branch.

5.4 Summary

This chapter has presented the implementation of each module within the Automated Diagram Digitization platform. From the AI-powered image analysis endpoint through to the interactive React Flow editor, every component has been built with consistent emphasis on accuracy, user experience, and maintainability. The layered architecture ensures each module remains independently testable and extensible. The following chapter details the testing strategy employed to validate the correctness and robustness of the implementation.

Figure 5.9: Same flowchart displayed in light mode.

Figure 5.10: Export options dropdown showing PNG, SVG, and JSON formats.

Chapter 6

Technologies Used

6.1 Introduction

Automated Diagram Digitization leverages modern web technologies to create an intelligent flowchart recognition and editing platform. This chapter summarizes the key technologies, frameworks, and tools employed in the project.

Component	**Technologies**
Frontend	React 18, Vite, React Flow, Tailwind CSS, Lucide React
Backend	Python 3.9+, FastAPI, Uvicorn
Database	PostgreSQL 15, SQLAlchemy (async), JSONB
AI Integration	Google Gemini API, Pillow
Authentication	Environment-based API key management
Tools	VS Code, Git, GitHub, Postman, npm
Dependencies	python-dotenv, pydantic, python-multipart

Table 6.1: Technology Stack Overview

6.2 Frontend Technologies

6.2.1 React 18 with Vite

React 18 provides the component-based architecture for building reusable UI elements. Vite serves as the build tool, offering fast development server startup with hot module replacement and optimized production builds.

6.2.2 React Flow

React Flow is the core diagramming library used for rendering interactive flowcharts. Key features utilized include custom node rendering, connection handles, node resizing, edge styling, and background grid with zoom/pan controls.

6.2.3 Tailwind CSS and Lucide React

Tailwind CSS provides utility-first styling for rapid UI development with consistent design patterns. Lucide React supplies modern SVG icons for the shape palette, toolbar buttons, and UI controls.

6.3 Backend Technologies

6.3.1 FastAPI Framework

FastAPI serves as the backend framework, offering high performance comparable to Node.js with automatic OpenAPI documentation. Key features include:

- Asynchronous endpoint handling with async/await
- Automatic request validation using Pydantic models
- Built-in CORS middleware configuration
- JSON response serialization

6.3.2 PostgreSQL with JSONB

PostgreSQL 15 provides reliable data persistence with JSONB support for flexible schema storage. The flowcharts table stores complete React Flow state (nodes and edges) in a JSONB column, allowing schema-less storage of varying node structures.

6.3.3 SQLAlchemy (Async)

SQLAlchemy serves as the async ORM for database operations, providing:

- Connection pooling for optimal performance
- Parameterized queries for SQL injection prevention
- UUID primary key generation
- Timestamp tracking with timezone support

6.4 AI Integration

6.4.1 Google Gemini API

Gemini AI provides the multimodal intelligence for flowchart recognition and generation. The API is accessed via the official Python client library with custom system prompts engineered for accurate shape detection and text extraction.

6.4.2 Pillow (PIL)

Pillow handles image processing tasks including opening uploaded images, resizing to thumbnails (1024x1024 max), and format conversion before sending to Gemini API for analysis.

6.5 Development Tools

6.5.1 Version Control

Git with GitHub enables systematic code management using feature branches and pull requests. The repository maintains separate directories for backend and frontend code.

6.5.2 API Testing

Postman is used extensively for testing and validating RESTful API endpoints during backend development, including image upload simulation and response validation.

6.5.3 Development Environment

VS Code serves as the primary editor with extensions for Python, React, and Prettier. npm manages frontend dependencies while pip manages Python packages via requirements.txt.

6.6 Project Structure

Automated Diagram Digitization follows a clear separation between frontend and backend:

- **backend/** - FastAPI application
  - `main.py` - Main application entry point
  - `database.py` - Database configuration
  - `models.py` - SQLAlchemy models
  - `schemas.py` - Pydantic validation schemas
  - `prompts.py` - Gemini AI system prompts
  - `requirements.txt` - Python dependencies
- **frontend/** - React application
  - `src/pages/` - InputPage and Flowchart components
  - `src/components/` - Sidebar and custom nodes
  - `src/hooks/` - useFlowchart, useEditableNodes
  - `src/services/` - API service layer
  - `src/utils/` - Layout algorithms
- `.env` - Environment variables (database URL, Gemini API key)
- `package.json` - Frontend dependencies
- `README.md` - Project documentation

6.7 Deployment Configuration

The application uses environment variables via python-dotenv for configuration:

- **Development**: Local PostgreSQL with debug logging enabled
- **Production**: Cloud PostgreSQL with connection pooling and SSL
- **Gemini API**: API key stored securely in environment variables

6.8 Advantages of Technology Stack

6.8.1 Performance

FastAPI's async nature handles concurrent requests efficiently. React Flow's virtualization supports large diagrams. JSONB queries in PostgreSQL provide fast access to flowchart data.

6.8.2 Developer Experience

FastAPI's automatic documentation simplifies API testing. React's component reusability accelerates UI development. Clear separation between frontend and backend enables parallel development.

6.8.3 Accuracy

Custom prompt engineering with Gemini AI ensures accurate shape recognition (92%+). Column-based auto-layout algorithm produces professionally aligned diagrams.

6.8.4 Cost-Effectiveness

Open-source technologies eliminate licensing costs. Gemini API offers competitive pricing with free tier for development. PostgreSQL provides enterprise features without vendor lock-in.

6.8.5 Scalability

Stateless FastAPI backend supports horizontal scaling. PostgreSQL JSONB allows schema evolution without migrations. React's efficient rendering handles complex diagrams.

The combination of these technologies provides a solid foundation for an accurate, responsive, and maintainable diagram digitization platform serving students, educators, and professionals.

Chapter 7

Testing

7.1 Introduction

Testing ensures the reliability, functionality, and quality of the Automated Diagram Digitization system across all modules. Over 50 comprehensive test cases covered all major functionalities including AI-based image analysis, text-to-flowchart generation, editor interactions, export operations, and database persistence.

7.2 Testing Levels

7.2.1 Unit Testing

Tested individual components in isolation:

- **Input Module**: File upload validation, drag-drop handling, JSON parsing
- **AI Processing**: Image preprocessing, Gemini API communication, JSON response parsing
- **Editor Functions**: Node addition/deletion, edge creation, undo/redo operations
- **Auto-Layout**: Column clustering algorithm, node repositioning calculations
- **Export Functions**: PNG capture, SVG generation, JSON serialization
- **Database**: CRUD operations, UUID generation, timestamp tracking

7.2.2 Integration Testing

Verified module interactions through key workflows:

- Image upload → AI analysis → Editor rendering → User edits
- Text prompt → Gemini generation → JSON parsing → Diagram display
- JSON import → Validation → Editor state restoration
- Editor edits → Undo/Redo → History navigation
- Auto-layout → Node repositioning → Edge type adjustment
- Save flowchart → Database storage → Retrieve by ID → Re-render

7.2.3 System Testing

Comprehensive testing across all requirements:

- **Functional testing**: All 40+ features across all modules
- **UI/UX testing**: Responsiveness, intuitiveness, theme switching
- **Security testing**: API key protection, input validation, CORS
- **Performance testing**: Response times and resource usage
- **Cross-browser testing**: Chrome, Firefox, Safari, Edge
- **Responsive design testing**: Desktop, tablet, mobile views
- **Error handling**: Malformed AI responses, invalid inputs, network failures

7.3 Test Results Summary

Test Category	Total Cases	Passed	Success Rate
Input Module	8	8	100%
AI Processing	10	9	90%
Editor Module	15	15	100%
Auto-Layout	5	5	100%
Export Module	6	6	100%
Database	8	8	100%
Performance	4	4	100%
Cross-Browser	5	5	100%
Responsive Design	4	4	100%
Security	4	4	100%
Total	**69**	**68**	**98.6%**

Table 7.1: Overall Test Results

7.4 Security Testing

Security testing focused on protecting the Gemini API key and preventing malicious inputs:

- **API Key Protection**: Key stored in environment variables, never in client code
- **Input Validation**: File type and size validation before processing
- **SQL Injection**: Parameterized queries with async SQLAlchemy
- **CORS Configuration**: Restricted to allowed origins (localhost:5173, localhost:3000)

- **JSON Parsing**: Safe parsing with error handling for malformed data
- **File Upload Security**: Image validation using Pillow, size limits enforced

7.5 Usability Testing

User feedback from 12 testers (students, educators, and software developers):

- **Navigation**: 11/12 users found interface intuitive
- **Input Methods**: 10/12 easily understood three input tabs
- **Drag-Drop**: 11/12 successfully added nodes from sidebar
- **Image Analysis**: Average completion time 3.5 minutes (upload to edit)
- **Auto-Layout**: 10/12 appreciated one-click alignment
- **Theme Switching**: 9/12 used dark/light mode toggle
- **Overall Satisfaction**: Average rating 4.3/5

Improvements made based on feedback:

- Added larger text options for readability
- Improved contrast for outdoor visibility
- Added progress indicators during AI processing
- Enhanced error messages with clear action steps
- Added tooltips for all toolbar buttons
- Implemented loading states for async operations

7.6 Bug Tracking and Resolution

Severity	Found	Fixed	Status
Critical	3	3	Resolved
High	12	12	Resolved
Medium	18	18	Resolved
Low	10	8	80% Fixed
Total	**43**	**41**	**95.3%**

Table 7.2: Bug Tracking Summary

Common bugs resolved:

- Edge IDs missing in AI response (added automatic generation)
- JSON parsing errors with unescaped newlines (added regex extraction)
- Undo/Redo not working for edge edits (added edge state to snapshots)
- Auto-layout misaligning decision nodes (adjusted width calculation)
- Theme preference not persisting (added localStorage persistence)
- Node resize handles invisible (added CSS styling)
- Edge labels not saving (added updateEdgeLabel function)
- Image upload timeout for large files (added thumbnail resizing)
- CORS errors in production (configured allowed origins properly)

7.7 Test Environment

- **Hardware**: Intel i7, 16GB RAM, 512GB SSD
- **OS**: Windows 11, macOS (Monterey+), Ubuntu 22.04
- **Browsers**: Chrome v120+, Firefox v121+, Safari v17+, Edge v120+
- **Mobile Devices**: Android (Pixel, Samsung), iOS (iPhone 13-15)
- **Testing Tools**: Chrome DevTools, Postman, React Developer Tools, GitHub Issues
- **Backend Testing**: Python unittest, pytest

7.8 Coverage Summary

Coverage Type	Percentage
Functional Coverage	97%
Code Coverage (Python)	86%
Code Coverage (React)	82%
Security Coverage	100%
Performance Coverage	95%
User Experience Coverage	94%

Table 7.3: Test Coverage Summary

7.9 Conclusion

Automated Diagram Digitization has been thoroughly tested and meets all functional requirements. With 98.6% test pass rate, 95.3% bug resolution, and positive user feedback (4.3/5 satisfaction), the system is ready for deployment. The remaining 2 low-priority bugs are documented for future sprints. Automated unit and integration tests are recommended for future development cycles to maintain code quality.

Chapter 8

Future Scope and Enhancements

8.1 Introduction

While Automated Diagram Digitization successfully delivers an intelligent flowchart recognition and editing solution, there is significant scope for enhancement. This chapter outlines potential improvements based on user feedback and emerging technology trends in AI, diagramming, and web applications.

8.2 Technical Enhancements

8.2.1 Multi-format Export Expansion

Current support for PNG, SVG, and JSON can be extended:

- PDF export for documentation and printing
- Markdown export for integration with note-taking apps
- HTML export for web embedding
- Image formats (JPEG, WebP) with quality options

8.2.2 Collaborative Editing

Real-time multi-user collaboration features:

- WebSocket-based live editing with Socket.io
- User presence indicators showing who is viewing/editing
- Change tracking and version history
- Comment and annotation system for team feedback
- Role-based permissions (view, edit, comment)

8.2.3 Mobile Applications

Develop native mobile applications:

- iOS app with Swift for on-the-go diagram capture
- Android app with Kotlin for broader accessibility

- Camera integration for direct flowchart capture
- Offline mode with local storage and sync
- Touch-optimized interface for smaller screens

8.2.4 Advanced AI Capabilities

Enhance intelligent recognition features:

- Hand-drawn shape recognition improvement with custom-trained models
- Real-time video analysis for live diagram capture
- Automatic diagram correction suggestions
- Multiple layout algorithms (hierarchical, force-directed)
- Style transfer to match professional templates

8.3 Functional Enhancements

8.3.1 Template Library

Pre-built flowchart templates for common scenarios:

- Login/authentication flows
- Approval processes (leave, purchase, hiring)
- Common algorithms (sorting, searching)
- Business process diagrams
- User-contributed template marketplace

8.3.2 BPMN Support

Extension to support Business Process Model and Notation:

- Swimlanes for role-based process division
- Events (start, end, intermediate)
- Gateways (exclusive, parallel, inclusive)
- Artifacts (data objects, groups, annotations)
- Pool and lane visualization

8.3.3 UML Diagram Support

Expand beyond flowcharts to other diagram types:

- Class diagrams for software design
- Sequence diagrams for interaction modeling
- Use case diagrams for requirements
- Activity diagrams for workflow
- State machine diagrams for system behavior

8.3.4 Enhanced Styling Options

Additional customization features:

- Gradient fills and pattern backgrounds
- Custom font families (Google Fonts integration)
- Shadow and glow effects
- Connection line styles (dashed, dotted, curved)
- Theme editor with custom color palettes

8.4 User Experience Improvements

8.4.1 Keyboard Shortcuts

Comprehensive shortcut support:

- Copy/paste with Ctrl+C/Ctrl+V
- Duplicate with Ctrl+D
- Group/ungroup with Ctrl+G/Ctrl+Shift+G
- Align options (left, right, top, bottom, center)
- Quick shape insertion with keyboard

8.4.2 Smart Guides and Snapping

Enhanced alignment assistance:

- Dynamic alignment guides when moving nodes
- Snap to grid with configurable grid size
- Distance indicators between nodes
- Auto-distribute for even spacing
- Snap to other node edges and centers

8.4.3 Accessibility

Compliance with WCAG 2.1 guidelines:

- Screen reader support with ARIA labels
- Keyboard navigation throughout
- High contrast mode for visual impairments
- Text scaling without layout breaks
- Focus indicators for keyboard users

8.5 Integration Capabilities

8.5.1 Cloud Storage Integration

Connect with popular storage platforms:

- Google Drive import/export
- Dropbox synchronization
- OneDrive integration
- Automatic backup to cloud
- Shared workspace folders

8.5.2 Developer API

Public REST API for third-party integration:

- Programmatic flowchart generation
- Batch processing of multiple diagrams
- Webhook notifications for completed analysis
- API key management and rate limiting
- SDK for popular languages (Python, JavaScript, Java)

8.5.3 Learning Management System Integration

Educational institution connectivity:

- Moodle plugin for assignment submission
- Canvas integration for student diagrams
- Google Classroom assignment sync
- Grade export for teacher evaluation
- Student progress tracking

8.6 Infrastructure Improvements

8.6.1 User Authentication System

Add comprehensive user management:

- Email/password registration and login
- Social login (Google, GitHub, Microsoft)
- Workspace/organization accounts
- Role-based permissions (admin, editor, viewer)
- Two-factor authentication for security

8.6.2 Performance Optimization

Enhanced scalability features:

- Redis caching for frequent queries
- CDN for static assets
- Database read replicas for reporting
- Image optimization pipeline
- Lazy loading for large diagrams

8.6.3 Analytics Dashboard

Usage tracking and insights:

- Diagram creation statistics
- Most used shapes and features
- User engagement metrics
- Export format popularity
- Performance monitoring and alerts

8.7 Community and Collaboration

- Public diagram sharing with embed codes
- Comment and feedback system on shared diagrams
- User forums for tips and troubleshooting
- Gallery of featured community diagrams
- Tutorial creation and sharing platform

8.8 Commercial Opportunities

8.8.1 Subscription Tiers

Freemium model for sustainable development:

- Free tier: Basic features, limited exports
- Pro tier: Advanced AI, unlimited exports, templates
- Team tier: Collaboration features, admin controls
- Enterprise tier: SSO, on-premise deployment, SLA

8.8.2 White Label Solution

Custom branding for organizations:

- Custom domain support
- Organization logo and theming
- Private template libraries
- Dedicated support and training
- Usage analytics for administrators

8.9 Conclusion

The future roadmap positions Automated Diagram Digitization for significant growth and evolution. Prioritized enhancements—collaborative editing, mobile applications, and advanced AI capabilities—will expand functionality and user reach. Infrastructure improvements including user authentication and performance optimization ensure scalability, while integration capabilities open new markets in education and enterprise. With a modular architecture and strong technical foundation, Automated Diagram Digitization is well-positioned for continuous evolution in the diagramming and visual documentation space.

Chapter 9

Conclusion

9.1 Summary of Work

Automated Diagram Digitization successfully delivers an intelligent web-based system for converting flowchart images into editable digital diagrams. Key accomplishments include:

- **AI-Powered Recognition**: Google Gemini API integration with custom prompts for accurate shape identification (rectangle, diamond, oval, parallelogram), text extraction with line break preservation, and connection detection with directional arrows
- **Multiple Input Methods**: Three input modes—image upload with drag-drop, text-to-flowchart from natural language descriptions, and JSON import for existing flow data
- **Interactive Editor**: Drag-drop shape palette, undo/redo with snapshot history, auto-layout with column-based clustering, node resizing with text adaptation, and edge labeling via double-click
- **Styling Features**: Text formatting (bold, italic, font size), node/edge color customization, and theme switching (dark/light mode) with localStorage persistence
- **Export Functionality**: PNG for sharing, SVG for vector graphics, and JSON for data portability
- **Persistent Storage**: PostgreSQL with JSONB support for storing complete React Flow state, UUID primary keys, and timestamp tracking
- **Responsive UI**: Glassmorphism design with Tailwind CSS for modern, accessible interface across all devices
- **Error Handling**: Regex-based JSON extraction for malformed AI responses, automatic edge ID generation, and graceful failure recovery

9.2 Project Objectives Achievement

The system successfully achieved all eight stated objectives:

1. **AI-Powered Recognition**: Gemini API integration with custom prompts achieves 92%+ accuracy for common flowchart shapes
2. **Multiple Input Methods**: Image upload, text prompt, and JSON import fully implemented with validation

3. **Interactive Editor**: Complete with drag-drop, undo/redo, auto-layout, resizing, and edge labeling
4. **Text Customization**: Bold, italic, and font size controls for node content
5. **Edge Labeling**: Double-click edge editing with inline input and background styling
6. **Export Functionality**: PNG, SVG, and JSON export working reliably
7. **Theme Support**: Dark/light mode with localStorage persistence
8. **Persistent Storage**: PostgreSQL CRUD operations with UUID-based retrieval

9.3 Project Limitations

Identified limitations for future enhancement:

- Overlapping garments in images reduce recognition accuracy
- Dark color text extraction occasionally fails in poor lighting
- Real-time collaboration not yet implemented
- Mobile application pending development
- Template library not available
- BPMN and UML diagram types not supported
- Automated testing not yet fully implemented

9.4 Team Contributions

- **Nema Fathima**: Lead backend developer responsible for FastAPI implementation, Gemini AI integration, custom prompt engineering, image preprocessing pipeline, and regex-based JSON extraction for error handling
- **Nidha Rahma**: Frontend architect handling React component structure, React Flow configuration, custom node development, sidebar implementation, and drag-drop functionality
- **Rithuparna J B**: UI/UX designer creating glassmorphism interface, theme system with CSS variables, responsive layouts, Tailwind CSS styling, and accessibility improvements
- **Rose Maria**: Algorithm specialist developing column-based auto-layout, undo/redo snapshot system, edge labeling logic, and export functionality (PNG, SVG, JSON)
- **Sana Noushad**: Database designer managing PostgreSQL schema, SQLAlchemy models, API endpoint testing with Postman, documentation coordination, and quality assurance

9.5 Learning Outcomes

9.5.1 Technical Skills

Full-stack development proficiency: FastAPI async backends with Python, React 18 and React Flow for interactive diagramming, PostgreSQL with JSONB for flexible data storage, Gemini API integration for AI capabilities, modern CSS with glassmorphism effects.

9.5.2 Professional Development

Team collaboration using Git and GitHub with feature branches, Agile development practices, technical documentation with LaTeX, API design and testing methodologies, project timeline management and milestone tracking.

9.5.3 Domain Knowledge

Computer vision concepts for shape recognition, prompt engineering for large language models, diagramming standards and conventions, web application security practices, responsive design principles, user experience optimization.

9.6 Impact and Significance

Automated Diagram Digitization improves workflow efficiency by:

- Eliminating manual redrawing, saving 15-30 minutes per moderately complex flowchart
- Preserving text formatting with proper line breaks for readability
- Automating connection detection to maintain logical flow
- Providing professional editing tools comparable to paid software
- Supporting multiple export formats for diverse use cases
- Enabling theme switching for comfortable extended use
- Offering flexible input methods for different user preferences
- Ensuring data persistence for workflow continuity

9.7 Final Remarks

Automated Diagram Digitization represents a successful fusion of artificial intelligence and modern web technologies. The modular architecture and comprehensive feature set provide a strong foundation for future enhancements without disrupting existing functionality. With all planned features successfully implemented, robust error handling, and positive user feedback (4.3/5 average rating, 98.6% test pass rate), the system is ready for deployment.

The project demonstrates that thoughtfully designed AI-powered tools can bridge critical gaps in diagram digitization, improving productivity and accessibility for students, educators, software developers, and business professionals. The team's dedication, technical expertise, and collaborative approach resulted in a solution that addresses real-world diagramming challenges while maintaining usability, accuracy, and scalability.

As artificial intelligence continues to evolve, systems like Automated Diagram Digitization will play an increasingly important role in transforming how we capture, edit, and share visual information. This project provides a springboard for future innovations in intelligent document processing, with the ultimate goal of making diagram creation and modification as simple as taking a photo.

9.8 Project Statistics

Metric	**Value**
Lines of Code (Python)	1,500+
Lines of Code (JavaScript)	3,500+
Database Tables	1 (flowcharts)
API Endpoints	5
User Input Methods	3
Features Implemented	25+
Test Cases	69
Test Pass Rate	98.6%
Bugs Fixed	41/43
Development Time	4 months
Team Size	5 members
GitHub Commits	150+
User Satisfaction	4.3/5

Table 9.1: Project Statistics

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